VIDYASHILP UNIVERSITY

Applied Deep Learning

(Semester-VII, 2026-27)

Project Work

Group-2 : Members

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Domain : e-Commerce
Project Title : Agentic AI based e-Commerce Assistant
Marks : 30
Issue Date : 16th October 2026.
Submission Deadline: 13th November 2026 (23.59 Hrs.).

A) Project Overview

Electronic commerce platforms handle thousands of customer enquiries relating to product selection, prices, product availability, delivery, returns, refunds, and order status. Traditional chatbots generally respond to predefined questions or generate answers based on fixed rules. They may not be capable of independently selecting and executing the sequence of actions needed to solve a customer's problem.

Agentic AI introduces a more flexible approach. An AI agent can interpret a customer's objective, reason about the task, select appropriate tools, retrieve information, perform permitted actions, and present a response based on the results.

The project requires the team to develop an e-Commerce AI Agent with limited capability using Python programming, Agentic AI Frameworks and a set of tools that the agent can invoke.

The said Agentic AI-based e-Commerce assistant is expected to be capable of understanding simple & very basic customer requests and carrying out selected e-Commerce tasks through Python-based tools.

The system should use a product catalogue and simulated e-Commerce business data to support intelligent product discovery and basic customer-service enquiries.

The agent must determine which tools are appropriate for a given request, invoke those tools with suitable parameters, interpret the returned information, and generate a final response.

B) Project objectives

The objective is to demonstrate how an AI agent can:

i) Select tools,
ii) Execute tasks,
iii) Observe results, and
iv) Use those results to accomplish a customer's goal.

C) Project Tasks

The limited scope of the proposed Agentic AI System is:

a) The agent identifies product requirements from a natural-language query and searches a product catalogue using Python tools.
b) The agent answers enquiries about simulated orders, delivery status, return policies, and refund procedures by calling appropriate tools.

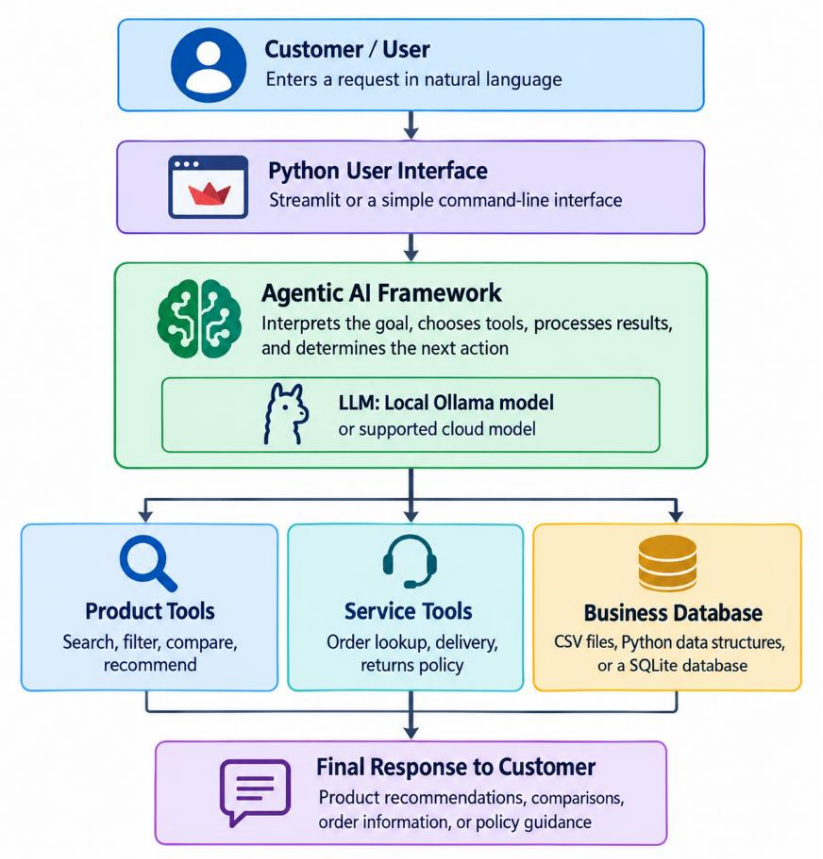
Note: However, the capability of the AI Agent can be expanded further by including more tools. But, the purpose of the project is to provide the students a chance to understand the Agentic AI approach and therefore, above-mentioned limited scope is adequate under the given time constraint.

The tasks to be done as part of this project are as follows:

1) Develop a Python-based AI agent using an existing LLM and an agentic framework (like LangChain or LlamaIndex).
2) Create a product catalogue containing product names, categories, prices, specifications, ratings, and stock availability.

3) Enable the agent to interpret customer requests expressed in natural language.
4) Develop Python tools that the agent can select and invoke to perform specific tasks.
5) Enable the agent to answer basic questions about orders, delivery, returns, and refunds using simulated business data.
6) Evaluate the system's accuracy. Use at least 20 test cases of Product Discovery and 20 test cases of Customer Service enquires.

D) Architecture of the AI Agent



E) Development Methodology

The team is required to complete the project in the following phases.

Phase 1: Data preparation

Create the product and order datasets.

Phase 2: Development of Tools in Python

Implement and test the product search, filtering, order lookup, and policy functions independently.

Phase 3: LLM and agent integration

Configure the chosen model, register the tools, and implement the agent workflow. Confirm that the agent can select the correct tools and supply valid arguments.

Phase 4: User interface development

Develop a simple Streamlit interface or command-line application through which a customer can submit queries and view responses.

Phase 6: Testing and evaluation

Execute test cases, record results and evaluate performance.

F) Dataset Requirements

The team is advised to create a small synthetic dataset or use a small publicly available e-Commerce dataset. A manageable dataset is preferable over a very large dataset because the emphasis is on developing an understanding of Agentic AI approach. The following datasets need to be created:

i) Product catalogue

Create a CSV file named products.csv with at least 30–50 products across three or more categories. Suggested fields are:

Field	Description	Example
product_id	Unique identifier	P101
product_name	Product name	Laptop A
category	Product category	Laptop
price	Selling price in rupees	54999
brand	Brand name	Brand X
rating	Average rating	4.3
stock	Units available	15
specifications	Key product features	16 GB RAM, 512 GB SSD

NOTE: The team should use clearly labelled fictional brands and products if creating synthetic data, or appropriately document the source and permitted use of any real product data.

ii) Order catalogue

Create orders.csv with at least 20 simulated orders. The suggested fields are:

- order_id
- customer_id
- product_id
- order_date
- order_status
- expected_delivery_date
- payment_status

The team can add a returns_policy.csv or a simple Python dictionary containing return period, eligibility conditions, and refund rules.

G) Evaluation Metrics

The team is required to evaluate the technical performance.

Metric	Evaluation approach
Tool-selection accuracy	Proportion of test requests for which the appropriate tool was selected.
Constraint satisfaction/Return Policy Compliance	Proportion of returned products meeting all specified requirements
Task completion rate	Proportion of test requests completed successfully

Example of Task Completion Rate

If the system is tested on 20 requests and successfully completes 17, the task completion rate is: Task Completion Rate=(17/20)*100=85%

H) Expected Deliverables

The team is required to submit the following:

1) Project report

Include the problem statement, objectives, literature/background review, architecture, implementation, testing and **limitations** of the AI Agent**.**

2) Python source code

Submit all source code files and setup instructions. **The source code must be well commented**. Please note that you must prepare setup instructions clearly. **The clarity of setup instructions must be such that the professor in-charge should be able to install the AI Agent independently without any assistance and run it successfully.**

3) Datasets

Include the product catalogue, order records, return policies, and a description of data sources or synthetic data generation.

4) Testing evidence

Provide the test-case sheet, results, screenshots or logs, evaluation metrics.

5) Presentation and live demonstration

Present the architecture, demonstrate at least three working scenarios, explain the tool-selection process, LLM configuration and briefly present the limitations.

I) Limitations & Scope for Expansion

This is an important part of the project. The team must clearly articulate the limitations (if any) of the AI Agent and also the scope for further expansion of the scope of the Agent. This must be covered in the Project Report.

J) Submission Guidelines

Put all your files like .pdf/.doc file, .py file, datasets etc. in a zip folder. Let the .zip folder be named as "Project-Group-x.zip" where x is your group number and submit it on Diji platform.

K) Evaluation Rubric

Evaluation Component	Marks	What is Assessed?
1. Report.	5	Good quality report clearly addressing all the aspects mentioned in section – H(1) of this document.
2. Source Code + Dataset	15	All items mentioned in Section-H(2) to H(4). The team must document clear and correct set-up instructions.

Evaluation Component	Marks	What is Assessed?
3. Presentation and interaction with professor in-charge	10	Presentation of the entire work by the team, performance of individual team members in the interaction with the professor in-charge.
Total	**30**	

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